PREVIOUS YEARS' QUESTIONS

30. What is the LCM of $a^3b - ab^3$, $a^3b^2 + a^2b^3$ and $ab(a+b)$? 2012 I
(a) $a^2b^2(a^2 - b^2)$ (b) $ab(a^2 - b^2)$
(c) $a^2b^2 + ab^3$ (d) $a^3b^3(a^2 - b^2)$

31. What is the HCF of $36(3x^4 + 5x^3 - 2x^2)$, $9(6x^3 + 4x^2 - 2x)$ and $54(27x^4 - x)$? 2012 I
(a) $9x(x+1)$ (b) $9x(3x-1)$ (c) $18x(3x-1)$ (d) $18x(x+1)$

32. What is the HCF of the polynomials $x^3 + 8$, $x^2 + 5x + 6$ and $x^3 + 2x^2 + 4x + 8$? 2013 II
(a) $x+2$ (b) $x+3$ (c) $(x+2)^2$ (d) None of these

33. The LCM of $(x^3 - x^2 - 2x)$ and $(x^3 + x^2)$ is 2013 II
(a) $x^3 - x^2 - 2x$ (b) $x^2 + x$ (c) $x^4 - x^3 - 2x^2$ (d) $x - 2$

34. The HCF of $(x^4 - y^4)$ and $(x^6 - y^6)$ is 2013 II
(a) $x^2 - y^2$ (b) $x - y$ (c) $x^3 - y^3$ (d) $x^4 - y^4$

35. What is the LCM of $x^2 + 2x - 8$, $x^3 - 4x^2 + 4x$ and $x^2 + 4x$? 2013 II
(a) $x(x+4)(x-2)^2$ (b) $x(x+4)(x-2)$
(c) $x(x+4)(x+2)^2$ (d) $x(x+4)^2(x-2)$

36. What is the HCF of $a^2b^4 + 2a^2b^2$ and $(ab)^7 - 4a^2b^9$? 2013 II
(a) ab (b) a^2b^3 (c) a^2b^2 (d) a^3b^2

37. What is the HCF of $8(x^5 - x^3 + x)$ and $28(x^6 + 1)$?
(a) $4(x^4 - x^2 + 1)$ (b) $2(x^4 - x^2 + 1)$ 2014 I
(c) $(x^4 - x^2 + 1)$ (d) None of these

38. What is the highest common factor of $2x^3 + x^2 - x - 2$ and $3x^3 - 2x^2 + x - 2$? 2014 II
(a) $x-1$ (b) $x+1$ (c) $2x+1$ (d) $2x-1$

39. The HCF and LCM of two polynomials are $(x+y)$ and $(3x^5 + 5x^4y + 2x^3y^2 - 3x^2y^3 - 5xy^4 - 2y^5)$, respectively. If one of the polynomials is $(x^2 - y^2)$, then the other polynomial is 2015 I
(a) $3x^4 - 8x^3y + 10x^2y^2 + 7xy^3 - 2y^4$
(b) $3x^4 - 8x^3y - 10x^2y^2 + 7xy^3 + 2y^4$
(c) $3x^4 + 8x^3y + 10x^2y^2 + 7xy^3 + 2y^4$
(d) $3x^4 + 8x^3y - 10x^2y^2 + 7xy^3 + 2y^4$

40. If $(x+1)$ is the HCF of $Ax^2 + Bx + C$ and $Bx^2 + Ax + C$ where $A \neq B$, then the value of C is 2015 II
(a) A (b) B (c) $A - B$ (d) 0

41. The sum and difference of two expressions are $5x^2 - x - 4$ and $x^2 + 9x - 10$ respectively. The HCF of the two expressions will be 2016 I
(a) $x+1$ (b) $(x-1)$ (c) $(3x+7)$ (d) $(2x-3)$

ANSWERS

1	c	2	b	3	c	4	a	5	b	6	a	7	a	8	a	9	d	10	b
11	b	12	d	13	c	14	a	15	b	16	a	17	b	18	c	19	d	20	a
21	c	22	b	23	c	24	a	25	a	26	c	27	c	28	b	29	c	30	a
31	c	32	a	33	c	34	a	35	a	36	c	37	a	38	a	39	c	40	d
41	b																		

HINTS AND SOLUTIONS

1. *(c)* LCM = product of the largest power of each factor

$$= x^2(x-1)(x-2)(x+3)$$

2. *(b)* Here, $2(a^2 - b^2) = 2(a+b)(a-b)$
$3(a^3 - b^3) = 3(a-b)(a^2 + ab + b^2)$
and $4(a^4 - b^4) = 4(a+b)(a-b)(a^2 + b^2)$

LCM of numerical coefficients = 12 and LCM of algebraic expressions

$$= (a-b)(a+b)(a^2+b^2)(a^2+ab+b^2)$$
$$= (a^4 - b^4)(a^2 + ab + b^2)$$

$\therefore$ LCM of polynomials

$$= 12(a^4 - b^4)(a^2 + ab + b^2)$$

3. *(c)* $\text{LCM} = \dfrac{\text{Product of expressions}}{\text{HCF}}$

$$= \frac{a \times b}{1} = ab$$

4. *(a)* Given, $(x+3)^2(x-2)(x+1)^2$ and $(x+1)^3(x+3)(x+4)$

LCM $= (x-2)(x+1)^3(x+3)^2(x+4)$

5. *(b)* $4y^4x - 9y^2x^3 = y^2x(4y^2 - 9x^2)$
$= y^2x(2y - 3x)(2y + 3x)$
$4y^2x^2 + 6yx^3 = 2yx^2(2y + 3x)$

$\therefore$ Required HCF $= xy(2y + 3x)$

6. *(a)* Let $p(x) = x^4 - x^2 - 6$
$= x^4 - 3x^2 + 2x^2 - 6$
$= x^2(x^2 - 3) + 2(x^2 - 3)$
$= (x^2 + 2)(x^2 - 3)$
$q(x) = x^4 - 4x^2 + 3$
$= x^4 - 3x^2 - x^2 + 3$
$= x^2(x^2 - 3) - 1(x^2 - 3)$
$= (x^2 - 3)(x^2 - 1)$

HCF of $p(x), q(x) = x^2 - 3$

7. *(a)* $A = (x+3)^2(x-2)(x+1)^2$ and $B = (x+1)^2(x+3)(x+4)$

$\therefore$ HCF of polynomials $= (x+3)(x+1)^2$

8. *(a)* Here, HCF of 22 and 36 is 2.

Now, $x(x+1)^2 = x(x+1)(x+1)$

$x^2(2x^2+3x+1) = x^2(2x+1)(x+1)$

Common factors of $x(x+1)^2$

and $x^2(2x^2+3x+1)$ are $x(x+1)$.

Hence, required HCF $= 2x(x+1)$

9. *(d)* $a^2 - b^2 - c^2 - 2bc = a^2 - (b^2 + c^2 + 2bc)$

$= a^2 - (b+c)^2 = (a+b+c)(a-b-c)$

$b^2 - c^2 - a^2 - 2ac = b^2 - (c^2 + a^2 + 2ac)$

$= b^2 - (c+a)^2$

$= (b-c-a)(b+c+a)$

$= (a+b+c)(b-c-a)$

and $c^2 - a^2 - b^2 - 2ab$

$= c^2 - (a^2 + b^2 + 2ab) = c^2 - (a+b)^2$

$= (c-a-b)(c+a+b)$

$= (a+b+c)(c-a-b)$

$\therefore$ Required LCM

$= (a+b+c)(a-b-c)$

$(b-c-a)(c-a-b)$

10. *(b)* Given, $[(x+3)(x-2)^2]$

and $[(x-2)(x-6)]$

$\therefore$ LCM $= (x+3)(x-2)^2(x-6)$

11. *(b)* Since, $(z-1)$ is the HCF, so it will divide each one of the given polynomials. So, $z = 1$ will make each one zero.

$\therefore \quad p(1)^2 - q(1+1) = 0 \Rightarrow p = 2q$

12. *(d)* Let $p(x)$ and $q(x)$ be two polynomials and $p(x) \div \text{HCF} = a$

and $q(x) \div \text{HCF} = b$

$\therefore \quad p(x) \times q(x) = \text{LCM} \times \text{HCF}$

$\Rightarrow a \times \text{HCF} \times b \times \text{HCF} = \text{LCM} \times \text{HCF}$

$\Rightarrow \quad a \times b \times \text{HCF} = \text{LCM}$

$$ab = \frac{\text{LCM}}{\text{HCF}}$$

$$= \frac{3x^4 + 4x^3 - 7x^2 - 4x + 4}{3x^2 + 4x - 4}$$

$= (x+1)(x-1)$

Let $a = (x+1)$ and $b = (x-1)$,

then the required expression are

$(x+1)(3x^2 + 4x - 4)$

and $(x-1)(3x^2 + 4x - 4)$.

13. *(c)* Here, LCM $= 36x^3(x+a)(x^3 - a^3)$ and HCF $= x^2(x-a)$

$p(x) = 4x^2(x^2 - a^2)$

But $p(x) \times q(x) = \text{HCF} \times \text{LCM}$

$$q(x) = \frac{\text{HCF} \times \text{LCM}}{p(x)}$$

$$= \frac{x^2(x-a)\,36x^3(x+a)(x^3 - a^3)}{4x^2(x^2 - a^2)}$$

$= 9x^3(x^3 - a^3)$

14. *(a)* Given,

$p(x) = x^2 + 2x - 3 = (x+3)(x-1)$

$q(x) = x^2 + x - 6 = (x+3)(x-2)$

and LCM

$= x^3 - 7x + 6 = (x-1)(x^2 + x - 6)$

$= (x-1)(x+3)(x-2)$

$$\therefore \text{HCF} = \frac{p(x) \times q(x)}{\text{LCM}}$$

$$= \frac{(x+3)(x-1) \times (x+3)(x-2)}{(x-1)(x+3)(x-2)}$$

$= (x+3)$

15. *(b)* Since, $(x+k)$ is the HCF, it will divide both the polynomials without leaving any remainder, thus $x = -k$ will make both of them zero.

$\therefore k^2 - pk + q = k^2 - ak + b$

or $\quad -ak + b = -pk + q$

$\Rightarrow \quad ak - pk = b - q$

$$\therefore \quad k = \frac{b-q}{a-p}$$

16. *(a)* Since, $(x+4)$ is HCF, so it will divide both the expressions i.e. $x = -4$ will make each one zero.

$\therefore \quad 2(-4)^2 + k(-4) - 12 = 0$

$\Rightarrow \quad 32 - 12 = 4k$

$\therefore \quad 20 = 4k \quad \Rightarrow \quad k = 5$

17. *(b)* Since, HCF of $x^2 + x - 12$ and $2x^2 - kx - 9$ is $(x-k)$, then $(x-k)$ will be the factor of $2x^2 - kx - 9$.

$\therefore \quad 2k^2 - k^2 - 9 = 0$

$\Rightarrow \quad k^2 - 9 = 0$

$\Rightarrow \quad k = \pm 3$

and factor of $x^2 + x - 12$ are $(x+4)(x-3)$.

Hence, value of k is 3.

18. *(c)* Here, $(x-3)$ is GCD, so is a factor of both of them.

$\therefore$ Putting $x = 3$, in both makes each polynomial zero.

$3^3 - 2(3)^2 + p(3) + 6 = 0 \Rightarrow p = -5$

$3^2 - 5(3) + q = 0 \Rightarrow q = 6$

$\therefore \quad 5q + 6p = 5(6) + (-5)(6)$

$= 30 - 30 = 0$

19. *(d)* Let the expressions be $p(x)$ and $q(x)$, then

$p(x) + q(x) = 5x^2 - x - 4 \quad \ldots$(i)

$p(x) - q(x) = x^2 + 9x - 10 \quad \ldots$(ii)

On adding Eqs. (i) and (ii), we have

$2p(x) = 6x^2 + 8x - 14$

$\Rightarrow \quad p(x) = 3x^2 + 4x - 7$

$= 3x^2 + 7x - 3x - 7$

$= x(3x+7) - 1(3x+7)$

$\therefore \quad p(x) = (3x+7)(x-1)$

Subtracting Eqs. (i) and (ii), we have

$2q(x) = 4x^2 - 10x + 6$

$\Rightarrow \quad q(x) = 2x^2 - 5x + 3$

$= 2x^2 - 3x - 2x + 3$

$= x(2x-3) - 1(2x-3)$

$= (x-1)(2x-3)$

$\therefore$ LCM of $p(x)$ and $q(x)$

$= (x-1)(2x-3)(3x+7)$

20. *(a)* $p(x) = (x+2)(x+1)(x^2 + 2x + a)$

$q(x) = (x+3)(x+4)(x^2 + 7x + b)$

As HCF is $(x+1)(x+3)$, then both $(x+1)$ and $(x+3)$ must be factors of $p(x)$ and $q(x)$.

For $p(x)$, $\quad (x+1)$ is already a factor, so $(x+3)$ must be a factor of $x^2 + 2x + a$.

So, $\quad (-3)^2 + 2(-3) + a = 0$

$\Rightarrow \quad 9 - 6 + a = 0$

$\therefore \quad a = -3$

For $q(x)$, $\quad (x+3)$ is already factor.

$\therefore (x+1)$ must be a factor of

$x^2 + 7x + b$.

$\therefore (-1)^2 + 7(-1) + b = 0$

$\therefore \quad b = 6$

So, $\quad a = -3$ and $b = 6$ is solution.

21. *(c)* Given, $f(x)$ and $g(x)$ vanish at $x = \frac{1}{2}$

So, $(2x-1)$ is a factor of $f(x)$ and $g(x)$ both.

Hence, HCF of $f(x)$ and $g(x) = 2x - 1$.

22. *(b)* Refer to question 13.

23. *(c)* As $(x-2)$ is the HCF of $(ax^2 + bx + c)$ and $(bx^2 + ax + c)$

So, it will divide both the expressions,

$\therefore \quad a(2)^2 + b(2) + c = 0$

$\Rightarrow \quad 4a + 2b + c = 0 \quad \ldots$(i)

and $b(2)^2 + a(2) + c = 0$

$\Rightarrow \quad 4b + 2a + c = 0 \quad \ldots$(ii)

adding Eqs. (i) and (ii), we get

$\Rightarrow 6a + 6b + 2c = 0$

$\Rightarrow \quad 2c = -6a - 6b$

$\Rightarrow \quad c = -3(a+b)$

24. *(a)* $A = p^2 + 8p + 12 = (p+2)(p+6)$

$B = p^2 + 2p - 24 = (p-4)(p+6)$

$C = p^2 + 15p + 54 = (p+9)(p+6)$

I. LCM of A, B

and $C = (p+2)(p+6)(p-4)(p+9)$

Thus, I is correct

II. HCF of A, B and $C = (p+6)$

Hence, II is incorrect.

25. *(a)* I. $x^2 - 6x + 9 = (x-3)(x-3)$

and $x^3 - 27 = x^3 - (3)^3$
$= (x-3)(x^2 + 3x + 9)$

$\therefore$ HCF $= x - 3$.

Hence, it is true.

II. LCM of $10x^2yz$, $15xyz$ and $20xy^2z^2$ is $60x^2y^2z^2$.

Hence, it is false.

III. $6x^2 - 7x - 3 = (2x-3)(3x+1)$

and $2x^2 + 11x - 21$
$= (x+7)(2x-3)$

Hence, HCF $= (2x-3)$, it is also true.

Hence, the statement I and III are correct.

26. *(c)* We know that, $(x+y)$ and $(x-y)$ are the factors of $(x^{10} - y^{10})$.

Hence, statements I and IV are true.

27. *(c)* We have, $A = pq - np = p(q-n)$ …(i)

$C = q^2 - 3nq + 2n^2$
$= q^2 - 2nq - nq + 2n^2$
$= q(q-2n) - n(q-2n)$
$= (q-n)(q-2n)$ …(ii)

$E = pq - np - mq + mn$
$= p(q-n) - m(q-n)$
$= (p-m)(q-n)$ …(iii)

$\therefore$ HCF of A, C and $E = (q-n)$

28. *(b)* We have, $E = (p-m)(q-n)$ … (i)

$D = pq - 2np - mq + 2mn$
$= p(q-2n) - m(q-2n)$
$= (p-m)(q-2n)$ …(ii)

$\therefore$ LCM of D and E
$= (p-m)(q-n)(q-2n)$

29. *(c)* The factors of given polynomials are as follows

$A = pq - np = p(q-n)$ …(i)

$B = pq - mq = q(p-m)$ …(ii)

$C = (q-n)(q-2n)$ …(iii)

$D = (p-m)(q-2n)$ …(iv)

$E = (p-m)(q-n)$ …(v)

There is no such factor which is common to all given five polynomials. Thus, HCF $(A, B, C, D, E) = 1$

30. *(a)* Here, $a^3b - ab^3 = ab(a^2 - b^2)$
$= ab(a-b)(a+b)$

$a^3b^2 + a^2b^3 = a^2b^2(a+b)$

$\therefore$ LCM $[(a^3b - ab^3), (a^3b^2 + a^2b^3), ab(a+b)]$
$= a^2b^2(a+b)(a-b)$
$= a^2b^2(a^2 - b^2)$

31. *(c)* Let $P(x) = 36(3x^4 + 5x^3 - 2x^2)$
$= 36x^2(3x^2 + 5x - 2)$
$= 36x^2(x+2)(3x-1)$

$Q(x) = 9(6x^3 + 4x^2 - 2x)$
$= 18x(3x^2 + 2x - 1)$
$= 18x(3x-1)(x+1)$

$R(x) = 54(27x^4 - x) = 54x(27x^3 - 1)$
$= 54x(3x-1)(9x^2 + 1 + 3x)$

HCF of $[36, 18, 54] = 18$

$\therefore$ HCF of $[P(x), Q(x), R(x)]$
$= 18x(3x-1)$

32. *(a)* Let $f(x) = x^3 + 8 = x^3 + 2^3$
$= (x+2)(x^2 - 2x + 4)$
$= (x+2)(x-2)(x-2)$

$g(x) = x^2 + 5x + 6$
$= x^2 + 3x + 2x + 6$
$= (x+3)(x+2)$

and $h(x) = x^3 + 2x^2 + 4x + 8$
$= (x+2)(x^2 + 4)$

$\therefore$ HCF of $\{f(x), g(x), h(x)\} = (x+2)$

33. *(c)* Let $f(x) = x^3 - x^2 - 2x$
$= x(x^2 - x - 2)$
$= x(x+1)(x-2)$

and $g(x) = x^3 + x^2$
$= x^2(x+1) = x \cdot x(x+1)$

$\therefore$ LCM of $[f(x), g(x)]$
$= x^2(x+1)(x-2)$
$= x^2(x^2 - x - 2)$
$= x^4 - x^3 - 2x^2$

34. *(a)* Let $f(x) = (x^4 - y^4)$
$= (x^2 - y^2)(x^2 + y^2)$
$= (x-y)(x+y)(x^2 + y^2)$

and $g(x) = (x^6 - y^6)$
$= (x^3 + y^3)(x^3 - y^3)$
$= (x+y)(x^2 - xy + y^2)(x-y)(x^2 + xy + y^2)$
$= (x-y)(x+y)(x^2 - xy + y^2)(x^2 + xy + y^2)$

$\therefore$ HCF of
$[f(x), g(x)] = (x-y)(x+y)$
$= x^2 - y^2$

35. *(a)* $x^2 + 2x - 8 = (x-2)(x+4)$

$x^3 - 4x^2 + 4x = x[x^2 - 4x + 4]$
$= x(x-2)^2$

$x^2 + 4x = x(x+4)$

Now, LCM of $(x^2 + 2x - 8)$, $(x^3 - 4x^2 + 4x)$ and $(x^2 + 4x)$
$= x(x-2)^2(x+4)$

36. *(c)* $a^2b^4 + 2a^2b^2 = a^2b^2(b^2 + 2)$ …(i)

and $(ab)^7 - 4a^2b^9 = a^7b^7 - 4a^2b^9$
$= a^2b^2(a^5b^5 - 4b^7)$ …(ii)

From Eqs. (i) and (ii), HCF $= a^2b^2$

37. *(a)* Let $p(x) = 8(x^5 - x^3 + x)$
$= 4 \times 2 \times x(x^4 - x^2 + 1)$

and $q(x) = 28(x^6 + 1)$
$= 7 \times 4[(x^2)^3 + (1)^3]$
$= 4 \times 7 \times (x^2 + 1)(x^4 - x^2 + 1)$

$\therefore$ HCF of $p(x)$ and $q(x)$
$= 4(x^4 - x^2 + 1)$

38. *(a)* Let $f(x) = 2x^3 + x^2 - x - 2$
$= (x-1)(2x^2 + 3x + 2)$

and $g(x) = 3x^3 - 2x^2 + x - 2$
$= (x-1)(3x^2 + x + 2)$

Hence, the highest factor of $f(x)$ and $g(x)$ is $(x-1)$.

39. *(c)* Given, HCF $= (x+y)$ and

LCM $= 3x^5 + 5x^4y + 2x^3y^2 - 3x^2y^3 - 5xy^4 - 2y^5$
$= x^3(3x^2 + 5xy + 2y^2) - y^3(3x^2 + 5xy + 2y^2)$
$= (3x^2 + 5xy + 2y^2)(x^3 - y^3)$

We know that,

Product of two polynomials
$=$ HCF $\times$ LCM

$\therefore$ Required polynomial

$$= \frac{(x+y)(x^3 - y^3)(3x^2 + 5xy + 2y^2)}{(x-y)(x+y)}$$

$$= \frac{(x-y)(x^2 + y^2 + xy)(3x^2 + 5xy + 2y^2)}{(x-y)}$$

$= (x^2 + y^2 + xy)(3x^2 + 5xy + 2y^2)$
$= 3x^4 + 8x^3y + 10x^2y^2 + 7xy^3 + 2y^4$

40. *(d)* $(x+1)$ is the HCF of $Ax^2 + Bx + C$ and $Bx^2 + Ax + C$

$\therefore$ $A(-1)^2 + B(-1) + C = 0$
$\Rightarrow$ $A - B + C = 0$
$\Rightarrow$ $C = B - A$

and $B(-1)^2 + A(-1) + C = 0$
$\Rightarrow$ $B - A + C = 0$
$\Rightarrow$ $C = A - B$
$\therefore$ $C = 0$

41. *(b)* Refer to question 19.

16

RATIONAL EXPRESSIONS

Usually (1-2) questions have been asked from this chapter. Generally questions are asked from this chapter are based on simplification of rational expressions.

RATIONAL EXPRESSIONS

An expression in the form of $\frac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are polynomials and $q(x) \neq 0$ is called a rational expression.

- In the rational expression $\frac{p(x)}{q(x)}$, $p(x)$ is called the numerator and $q(x)$ is called the denominator of the rational expression.
- Every polynomial is a rational expression. Since, $p(x)$ can always be written as $\frac{p(x)}{1}$ and a constant function 1 is a polynomial of degree 0.
- Every rational expression need not be a polynomial.

Working Rule to Reduce the Given Rational Expression in its Lowest Term

1. Firstly, factorize both the polynomials $p(x)$ and $q(x)$.
2. Find the HCF of $p(x)$ and $q(x)$. If HCF of $p(x)$ and $q(x)$ is one, then rational expression $\frac{p(x)}{q(x)}$ is in its lowest terms.
3. If HCF is not equal to 1. Then, divide both $p(x)$ and $q(x)$ by their HCF and the rational expression is obtained in the lowest terms.

EXAMPLE 1. The lowest term of an expression $\frac{a^3 - b^3}{a^2 + ab + b^2}$ is

a. $a+b$ b. $a-b$ c. ab d. $\frac{a}{b}$

Sol. b. Rational expression

$$= \frac{a^3 - b^3}{a^2 + ab + b^2} = \frac{(a-b)(a^2 + ab + b^2)}{(a^2 + ab + b^2)} = a - b$$

$$[\because (a^3 - b^3) = (a-b)(a^2 + ab + b^2)]$$

EXAMPLE 2. The lowest term of an expression $\frac{3x^2 - 11x - 4}{6x^2 - 7x - 3}$ is

a. $\frac{x+4}{2x+3}$ b. $\frac{x+4}{2x-3}$ c. $\frac{x-4}{2x+3}$ d. $\frac{x-4}{2x-3}$

Sol. d. $\frac{3x^2 - 11x - 4}{6x^2 - 7x - 3} = \frac{3x^2 - 12x + x - 4}{6x^2 - 9x + 2x - 3}$

$$= \frac{3x(x-4) + 1(x-4)}{3x(2x-3) + 1(2x-3)} = \frac{(3x+1)(x-4)}{(3x+1)(2x-3)} = \frac{x-4}{2x-3}$$

EXAMPLE 3. The lowest term of an expression $\frac{12x^3y^5z^4}{18x^2y^6z^5}$ is

a. $\frac{2x}{yz}$ b. $\frac{xy}{z}$ c. $\frac{2x}{3yz}$ d. $\frac{3x}{2yz}$

Sol. c. $\frac{12x^3y^5z^4}{18x^2y^6z^5} = \frac{6 \times 2 \times x^2 \times x \times y^5 \times z^4}{6 \times 3 \times x^2 \times y \times y^5 \times z \times z^4} = \frac{2x}{3yz}$

Operations on Rational Expressions

1. **Addition** or **Subtraction of Rational Expressions with like denominators**

   It $\dfrac{P(x)}{q(x)}$ and $\dfrac{h(x)}{q(x)}$ are two rational expressions then

   $\dfrac{p(x)}{q(x)} \pm \dfrac{h(x)}{q(x)} = \dfrac{p(x) \pm h(x)}{q(x)}$

2. **Addition** or **subtraction of Rational Expressions with unlike denominators**

To add or subtract rational expressions with unlike denominators, follow the steps given below

Step I Write each denominator in the factor form

Step II Find the LCM of the denominators

Step III Rewrite each rational expression with LCM as the denominator

Step IV Add or subtract the numerators.

EXAMPLE 4. The sum of $\dfrac{x+1}{x-1}$ and $\dfrac{x-1}{x+1}$ is

a. $\dfrac{x^2+1}{x^2-1}$ b. $\dfrac{1}{x^2+1}$ c. $\dfrac{2x^2+2}{x^2-1}$ d. $\dfrac{x^2+2}{x^2-1}$

Sol. c. $\dfrac{x+1}{x-1} + \dfrac{x-1}{x+1} = \dfrac{(x+1)}{(x-1)} \times \dfrac{(x+1)}{(x+1)} + \dfrac{(x-1)}{(x+1)} \times \dfrac{(x-1)}{(x-1)}$

$= \dfrac{(x+1)^2 + (x-1)^2}{(x^2-1)}$

$= \dfrac{x^2+1+2x+x^2+1-2x}{x^2-1} = \dfrac{2x^2+2}{x^2-1}$

$[\because (a+b)^2 = a^2+b^2+2ab \text{ and } (a-b)^2 = a^2+b^2-2ab]$

EXAMPLE 5. What should be subtracted from $\dfrac{7x}{x^2+x-12}$ to get $\dfrac{4}{x+4}$?

a. $\dfrac{1}{x-3}$ b. $\dfrac{3}{x-3}$ c. $\dfrac{1}{x+3}$ d. $\dfrac{3}{x+3}$

Sol. b. Let $p(x)$ is to be subtracted, then

$\dfrac{7x}{x^2+x-12} - p(x) = \dfrac{4}{x+4}$

$\Rightarrow \quad p(x) = \dfrac{7x}{x^2+x-12} - \dfrac{4}{x+4} = \dfrac{7x}{(x+4)(x-3)} - \dfrac{4}{(x+4)}$

$= \dfrac{7x}{(x+4)(x-3)} - \dfrac{4(x-3)}{(x+4)(x-3)} = \dfrac{7x-4(x-3)}{(x+4)(x-3)}$

$= \dfrac{3x+12}{(x+4)(x-3)} = \dfrac{3(x+4)}{(x+4)(x-3)} = \dfrac{3}{x-3}$

3. **Multiplication of Rational Expressions If** $\dfrac{p(x)}{q(x)}$ and $\dfrac{g(x)}{h(x)}$ are two rational expressions, then their product is given by $\dfrac{p(x)}{q(x)} \times \dfrac{g(x)}{h(x)} = \dfrac{p(x) \cdot g(x)}{q(x) \cdot h(x)}$

   - Multiplicative inverse of $\dfrac{p(x)}{q(x)}$ is $\dfrac{q(x)}{p(x)}$.
   - 1 is the multiplicative identity.

EXAMPLE 6. The product of $\dfrac{x^2-1}{x^2+1}$ and $\dfrac{x+2}{x+1}$ is

a. $\dfrac{(x-1)}{(x^2+1)}$ b. $\dfrac{(x-2)}{(x^2+1)}$

c. $\dfrac{(x-1)(x+2)}{(x^2+1)}$ d. $\dfrac{x+2}{(x^2-1)}$

Sol. c. Here, product $= \dfrac{x^2-1}{x^2+1} \times \dfrac{(x+2)}{(x+1)} = \dfrac{(x-1)(x+1)(x+2)}{(x^2+1)(x+1)}$

$[\because a^2-b^2 = (a-b)(a+b)]$

$= \dfrac{(x-1)(x+2)}{(x^2+1)}$

4. **Division of Rational Expressions** If $\dfrac{p(x)}{q(x)}$ and $\dfrac{g(x)}{h(x)}$ are two rational expressions, then their division is given by $\dfrac{p(x)}{q(x)} \div \dfrac{g(x)}{h(x)} = \dfrac{p(x)}{q(x)} \times \dfrac{h(x)}{g(x)}$

   = Product of $\dfrac{p(x)}{q(x)}$ and the reciprocal of $\dfrac{g(x)}{h(x)}$ i.e. $\dfrac{h(x)}{g(x)}$.

EXAMPLE 7. The lowest term of an expression $\dfrac{x^2+8x+12}{x^2-7x+12} \div \dfrac{x^2+4x-12}{x-4}$ is

a. $\dfrac{(x-2)}{(x-3)(x+2)}$ b. $\dfrac{x-2}{x+2}$

c. $\dfrac{(x+2)}{(x-3)(x-2)}$ d. $\dfrac{x+3}{x-2}$

Sol. c. Here, $\dfrac{x^2+8x+12}{x^2-7x+12} \div \dfrac{x^2+4x-12}{x-4}$

$= \dfrac{x^2+6x+2x+12}{x^2-4x-3x+12} \times \dfrac{x-4}{(x^2+6x-2x-12)}$

$= \dfrac{(x+6)(x+2)}{(x-4)(x-3)} \times \dfrac{x-4}{(x+6)(x-2)} = \dfrac{x+2}{(x-3)(x-2)}$